

GCSE algebra - Set 1 - Solutions

Specification M4.1-M4.8, M4.18, M4.19.

Answer key

Question	Answer	Topic
1	E	GCSE algebra
2	A	GCSE algebra
3	A	GCSE algebra
4	C	GCSE algebra
5	A	GCSE algebra
6	D	GCSE algebra
7	C	GCSE algebra
8	E	GCSE algebra
9	F	GCSE algebra
10	F	GCSE algebra
11	C	GCSE algebra
12	C	GCSE algebra

Worked solutions

Question 1 ALG-01 · GCSE algebra · spec M4.7

Make x the subject of $y = \frac{2x+3}{x-1}$.

Answer: E

$$y(x-1) = 2x+3 \Rightarrow xy-2x = y+3 \Rightarrow x(y-2) = y+3 \Rightarrow x = \frac{y+3}{y-2}$$

Full method - how and why

Rearranging when the new subject appears **twice**. The key move is collecting both x terms on one side and factorising.

1 Clear the fraction: multiply both sides by $(x-1)$: $y(x-1) = 2x+3$.

why: Get rid of the denominator first; everything else is then ordinary algebra.

2 Expand: $xy - y = 2x + 3$.

why: Multiply through the bracket.

3 Collect the x terms on one side: $xy - 2x = y + 3$.

why: Both terms containing x must end up together - this is the step the question is testing.

4 Factorise and divide: $x(y-2) = y+3$, so $x = \frac{y+3}{y-2}$.

why: Factoring out x turns two occurrences into one, which can then be divided out.

Where it goes wrong

- Dividing by y too early, leaving x on both sides.
- Sign slip in collecting, giving $\frac{y+3}{2-y}$ - an offered option.

Takeaway When the subject appears twice: expand, gather its terms, factorise, divide.

Question 2 ALG-02 · GCSE algebra · spec M4.19

Find the n th term of the quadratic sequence 3, 8, 15, 24, 35, ...

Answer: A

Second differences are 2, so the n th term is $n^2 + bn + c$: $n = 1$ gives $1 + b + c = 3$ and $n = 2$ gives $4 + 2b + c = 8$. Subtracting, $3 + b = 5$, so $b = 2$ and $c = 0$: $n^2 + 2n = n(n+2)$.

Full method - how and why

A quadratic sequence. The second difference identifies the n^2 coefficient; two terms then fix the rest.

1 First differences: 5, 7, 9, 11. Second differences: 2 (constant).

why: Constant second differences mean the sequence is quadratic.

2 The n^2 coefficient is half the second difference: $\frac{2}{2} = 1$, so the n th term is $n^2 + bn + c$.

why: For $an^2 + bn + c$, the second difference is always $2a$.

3 Use $n = 1$: $1 + b + c = 3$; and $n = 2$: $4 + 2b + c = 8$.

why: Two equations for the two remaining unknowns.

4 Subtract: $3 + b = 5$, so $b = 2$ and then $c = 0$: the n th term is $n^2 + 2n$.

why: Check $n = 4$: $16 + 8 = 24$ ✓.

Where it goes wrong

- Using the second difference itself as a , giving $2n^2 + \dots$
- Checking only the first term, which several options match.

Takeaway Second difference $= 2a$; then substitute two terms and verify with a third.

Question 3 ALG-04 · GCSE algebra · spec M4.8

Which of the following is an identity (true for all values of x)?

Answer: A

$$(x+3)^2 - (x-3)^2 = (x^2 + 6x + 9) - (x^2 - 6x + 9) = 12x \text{ for all } x.$$

Full method - how and why

An identity holds for **every** x . Expand each candidate and compare, or test values to eliminate.

- 1 Expand the first option: $(x+3)^2 - (x-3)^2 = (x^2 + 6x + 9) - (x^2 - 6x + 9)$.

why: Expand both squares fully, including the cross terms.

- 2 Simplify: the x^2 and 9 cancel, leaving $12x$ - true for all x . **This is the identity.**

why: Both sides agree identically, which is what 'identity' requires.

- 3 Check the others quickly: $(x+3)^2 = x^2 + 6x + 9 \neq x^2 + 9$; $(x+3)(x-3) = x^2 - 9$, not $+9$; $x^2 + 6x + 9 = 0$ holds only at $x = -3$; $(x-3)^2 = x^2 - 6x + 9$, not -9 .

why: Each wrong option drops or mis-signs a term - a quick test at $x = 1$ kills them all.

Where it goes wrong

- Confusing an equation (true sometimes) with an identity (true always).
- Forgetting the middle term when squaring a bracket.

Takeaway $a^2 - b^2 = (a-b)(a+b)$ applied to brackets gives quick identities; test $x = 1$ to eliminate impostors.

Question 4 ALG-05 · GCSE algebra · spec M4.18

A car accelerates uniformly from rest to 20 m/s in 8 seconds, travels at 20 m/s for 12 seconds, then decelerates uniformly to rest in 10 seconds. Find the total distance travelled.

Answer: C

$$\text{Area under the speed-time graph: } \frac{1}{2} \cdot 8 \cdot 20 + 12 \cdot 20 + \frac{1}{2} \cdot 10 \cdot 20 = 80 + 240 + 100 = 420 \text{ m.}$$

Full method - how and why

On a speed-time graph, distance is the **area** underneath. Split the shape into simple pieces.

- 1 Acceleration phase: a triangle of base 8 s and height 20 m/s: area $\frac{1}{2}(8)(20) = 80$ m.

why: Uniform acceleration from rest gives a straight line from the origin, so the region is a triangle.

- 2 Constant-speed phase: a rectangle $12 \times 20 = 240$ m.

why: Constant speed gives a horizontal line.

- 3 Deceleration phase: a triangle $\frac{1}{2}(10)(20) = 100$ m.

why: Uniform deceleration to rest gives another triangle.

- 4 Add: $80 + 240 + 100 = 420$ m.

why: Total area under the whole graph.

Where it goes wrong

- Treating the accelerating phases as rectangles, giving 600 m - an offered option.
- Using the gradient (acceleration) rather than the area.

Takeaway Speed-time graph: area = distance, gradient = acceleration.

Question 5 ALG-06 · GCSE algebra · spec M4.3

Given that $v^2 = u^2 + 2as$, find s when $v = 13$, $u = 5$ and $a = 4$.

Answer: A

$$169 = 25 + 8s \Rightarrow s = 18.$$

Full method - how and why

Substitution into a formula. Evaluate the squares first, then solve the linear equation.

1 Substitute: $13^2 = 5^2 + 2(4)s$, i.e. $169 = 25 + 8s$.

why: Compute each square separately; $2as = 2 \times 4 \times s = 8s$.

2 Rearrange: $8s = 144$.

why: $169 - 25 = 144$.

3 Divide: $s = 18$.

why: $144 \div 8 = 18$.

Where it goes wrong

- Forgetting the factor 2 in $2as$ and getting 36 - an offered option.
- Computing $13^2 = 139$ or $5^2 = 10$.

Takeaway Substitute numbers for every letter, evaluate powers first, then solve.

Question 6 ALG-07 · GCSE algebra · spec M4.6

Simplify $\frac{3}{x+2} - \frac{2}{x-1}$.

Answer: D

$$\frac{3(x-1) - 2(x+2)}{(x+2)(x-1)} = \frac{x-7}{(x+2)(x-1)}.$$

Full method - how and why

Subtracting algebraic fractions: common denominator, then expand the numerator carefully - especially the minus sign.

1 Common denominator: $(x+2)(x-1)$.

why: The denominators share no factor, so their product is the lowest common denominator.

2 Rewrite each fraction: $\frac{3(x-1)}{(x+2)(x-1)} - \frac{2(x+2)}{(x+2)(x-1)}$.

why: Multiply each numerator by whatever its denominator was missing.

3 Combine and expand: $3x - 3 - (2x + 4) = 3x - 3 - 2x - 4 = x - 7$.

why: The minus sign applies to **both** terms of the second bracket - the classic error.

4 So the answer is $\frac{x-7}{(x+2)(x-1)}$.

why: Numerator does not factorise further.

Where it goes wrong

- Getting $x+1$ by only negating the $2x$ - an offered option.
- Subtracting the denominators as well.

Takeaway Bracket the whole second numerator before subtracting, then distribute the minus sign.

Question 7 ALG-09 · GCSE algebra · spec M4.7

Given that $T = 2\pi\sqrt{\frac{l}{g}}$, express l in terms of T and g .

Answer: C

$$\frac{T}{2\pi} = \sqrt{\frac{l}{g}} \Rightarrow \frac{T^2}{4\pi^2} = \frac{l}{g} \Rightarrow l = \frac{gT^2}{4\pi^2}.$$

Full method - how and why

Rearranging a formula containing a square root. Isolate the root, square, then solve.

- 1 Isolate the square root: $\frac{T}{2\pi} = \sqrt{\frac{l}{g}}$.
why: Divide both sides by 2π before squaring, so the 2π gets squared correctly.
- 2 Square both sides: $\frac{T^2}{4\pi^2} = \frac{l}{g}$.
why: $(2\pi)^2 = 4\pi^2$ - squaring a product squares every factor.
- 3 Multiply by g : $l = \frac{gT^2}{4\pi^2}$.
why: g was dividing l , so it multiplies across.

Where it goes wrong

- Squaring before isolating, leaving 2π unsquared and giving $\frac{T^2 g}{2\pi}$ - an offered option.
- Putting g in the denominator, giving $\frac{T^2}{4\pi^2 g}$.

Takeaway Isolate the root first, then square everything on that side - constants included.

Question 8 ALG-10 · GCSE algebra · spec M4.18

A ball is thrown vertically upwards and its height after t seconds is $h = 20t - 5t^2$ metres. For how long is the ball at least 15 m above the ground?

Answer: E

$$20t - 5t^2 \geq 15 \Rightarrow t^2 - 4t + 3 \leq 0 \Rightarrow 1 \leq t \leq 3: \text{two seconds.}$$

Full method - how and why

A quadratic inequality in a real context. Solve it, then answer the question about **duration**.

- 1 Set up: $20t - 5t^2 \geq 15$.
why: 'At least 15 m' means the height is greater than or equal to 15.
- 2 Rearrange and divide by -5 (reversing the inequality): $-5t^2 + 20t - 15 \geq 0 \Rightarrow t^2 - 4t + 3 \leq 0$.
why: Dividing an inequality by a negative number flips its direction.
- 3 Factorise: $(t - 1)(t - 3) \leq 0$, so $1 \leq t \leq 3$.
why: Between the roots for ≤ 0 .
- 4 The **duration** is $3 - 1 = 2$ seconds.
why: The question asks how long, not when - a common misread.

Where it goes wrong

- Answering 3 (the later time) or 1 (the earlier time) instead of the interval length.
- Forgetting to flip the inequality when dividing by -5 .

Takeaway Solve for the interval, then read whether the question wants a time or a length of time.

Question 9 ALG-11 · GCSE algebra · spec M4.4

Expand and simplify $(x + 2)(x - 1)(x + 3)$.

Answer: F

$$(x^2 + x - 2)(x + 3) = x^3 + 3x^2 + x^2 + 3x - 2x - 6 = x^3 + 4x^2 + x - 6.$$

Full method - how and why

Expanding three brackets: do two at a time, then multiply the result by the third.

- 1 Multiply the first two: $(x + 2)(x - 1) = x^2 + x - 2$.

why: $2x - x = x$ for the middle term.

- 2 Multiply by $(x + 3)$: $(x^2 + x - 2)(x + 3) = x^3 + 3x^2 + x^2 + 3x - 2x - 6$.

why: Every term of the first bracket multiplies every term of the second - six products.

- 3 Collect: $x^3 + 4x^2 + x - 6$.

why: $3x^2 + x^2 = 4x^2$ and $3x - 2x = x$.

Where it goes wrong

- Sign slip on the x term, giving $x^3 + 4x^2 - x - 6$ - an offered option.
- Missing one of the six products.

Takeaway Two brackets at a time, and check the constant term ($2 \times -1 \times 3 = -6$) as a quick verification.

Question 10 ALG-12 · GCSE algebra · spec M4.19

A sequence has first term 2 and each subsequent term is obtained by squaring the previous term and subtracting 2. What is the 50th term?

Answer: F

$$2^2 - 2 = 2: \text{ every term is } 2.$$

Full method - how and why

A recurrence with a fixed point. Computing one term reveals the sequence never changes.

- 1 Apply the rule to the first term: $u_2 = 2^2 - 2 = 2$.

why: 'Square then subtract 2' applied to 2 returns 2.

- 2 Since $u_2 = u_1$, the same calculation repeats forever: every term is 2.

why: 2 is a fixed point of the map $x \mapsto x^2 - 2$.

- 3 So $u_{50} = 2$.

why: No need to iterate fifty times.

Where it goes wrong

- Assuming the terms must grow because of the squaring, and choosing 2^{50} .
- Computing $2^2 - 2$ as 0 or -2 .

Takeaway Always compute the second term: if it repeats the first, the sequence is constant.

Question 11 ALG-13 · GCSE algebra · spec M4.5

Factorise $6x^2 - 7x - 20$.

Answer: C

$$(3x + 4)(2x - 5) = 6x^2 - 15x + 8x - 20 = 6x^2 - 7x - 20.$$

Full method - how and why

Factorising $ax^2 + bx + c$ with $a \neq 1$. Use the 'split the middle term' method, or expand the options.

- 1 Multiply a and c : $6 \times (-20) = -120$. Find two numbers with product -120 and sum -7 : -15 and 8 .

why: This is the standard method; the pair always exists when the quadratic factorises over the integers.

- 2 Split the middle term: $6x^2 - 15x + 8x - 20$.

why: Order does not matter.

- 3 Factor in pairs: $3x(2x - 5) + 4(2x - 5) = (3x + 4)(2x - 5)$.

why: The bracket must match in both halves - that is the check that the split was right.

- 4 Verify by expanding: $6x^2 - 15x + 8x - 20 = 6x^2 - 7x - 20$ ✓.

why: Expanding the chosen option is the fastest check in a multiple-choice setting.

Where it goes wrong

- Choosing $(3x - 5)(2x + 4)$, which has a common factor 2 in the second bracket - a sign it cannot be right for this expression.
- Getting the signs the wrong way round: $(3x - 4)(2x + 5)$ gives $+7x$.

Takeaway Split the middle term using two numbers with product ac and sum b ; then expand your answer to check.

Question 12 ALG-14 · GCSE algebra · spec M4.16

The sum of two numbers is 12 and the difference of their squares is 48. Find the larger number.

Answer: C

$$x^2 - y^2 = (x - y)(x + y) = 12(x - y) = 48 \Rightarrow x - y = 4. \text{ So } x = 8, y = 4.$$

Full method - how and why

A sum and a difference of squares. Factorising the difference of squares links them without solving for both numbers.

- 1 Factorise: $x^2 - y^2 = (x - y)(x + y)$.

why: This is the identity that connects the two given facts.

- 2 Substitute the known sum: $48 = (x - y)(12)$, so $x - y = 4$.

why: The sum $x + y = 12$ is given directly.

- 3 Solve the pair $x + y = 12$, $x - y = 4$: adding gives $2x = 16$, so $x = 8$ (and $y = 4$).

why: Adding the two equations eliminates y .

- 4 The larger number is 8.

why: Check: $64 - 16 = 48$ ✓.

Where it goes wrong

- Answering 4 (the smaller number or the difference).
- Trying to solve $x^2 - y^2 = 48$ by trial.

Takeaway $x^2 - y^2 = (x - y)(x + y)$ converts a difference of squares into a linear pair.